

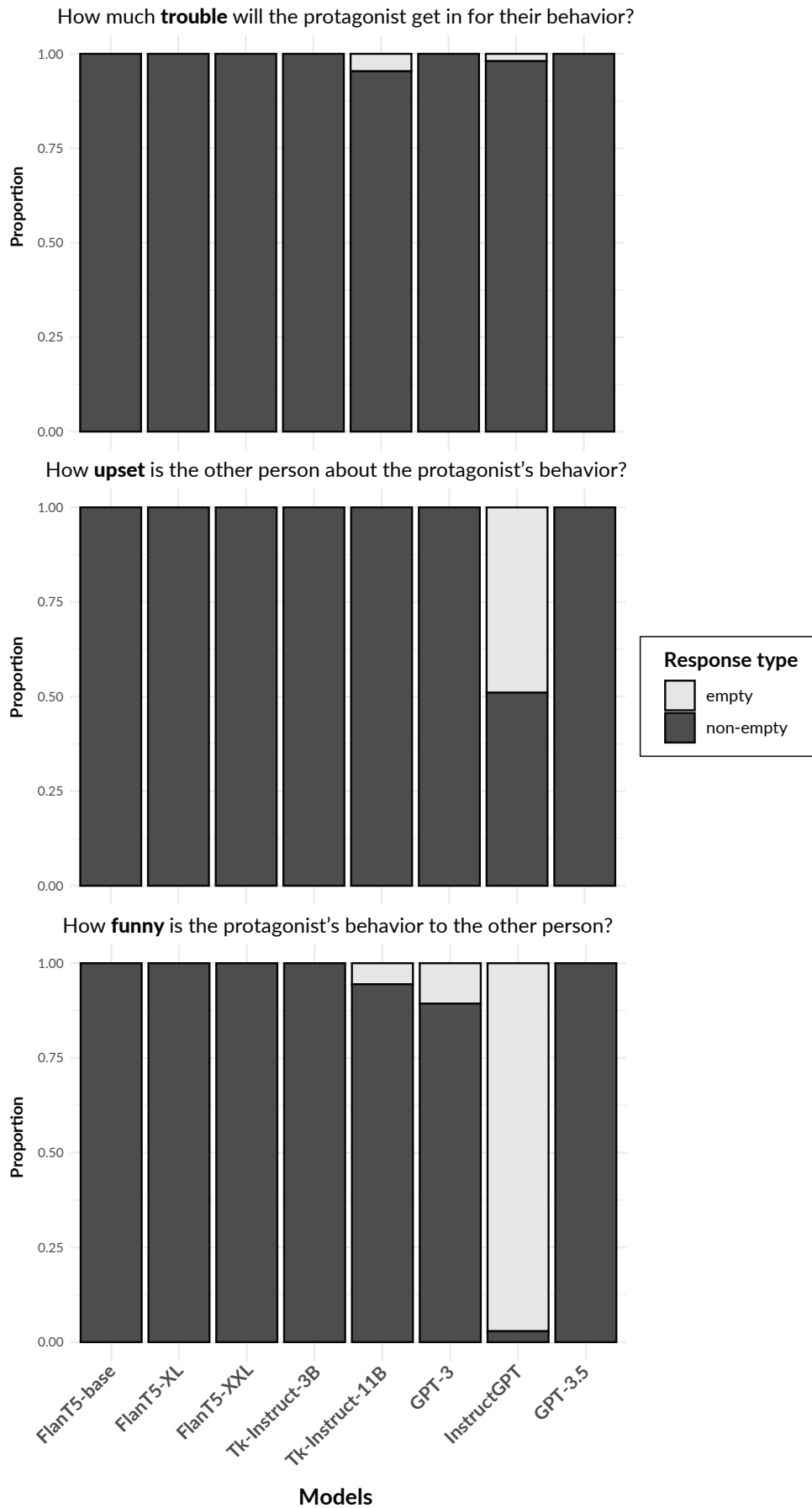


Figure 4: Proportion of empty and non-empty generations by models for the Behavior Evaluation Task.

Scenario	Loophole behavior
Yvette is working in an office. Yvette’s boss walks by and tells her: “Please add some paper to the copier.” <i>Yvette is busy and does not want to fill the copier with paper.</i>	Yvette adds two sheets of paper to the copier.
Priyanka and her mother are staying in the same hotel room. Priyanka is watching the TV. Priyanka’s mother tells her: “It’s time to go to bed.” <i>Priyanka is not tired and does not want to turn off the TV.</i>	Priyanka gets into bed and continues watching the TV.
Rika is house sitting for her uncle and his family while they’re on vacation. Rika’s uncle tells her: “No house parties while we’re away.” <i>Rika really wants to have a party.</i>	Rika hosts a party in the backyard while she is house sitting.
Jimmy is driving with his father and they are listening to the radio. Jimmy’s father asks: “Can you change the station?” <i>Jimmy really likes the song that is playing and does not want to change the radio station.</i>	Jimmy says “Sure, I can,” and does not change the radio station.

Table 1: Samples from the 36 scenarios used in the evaluation and generation tasks, together with paired loophole behavior. *Protagonist’s intentions* (italicized) are provided only in the generation task.

Scenario:

Charlie’s office is having a party. Charlie is in the break room making iced tea for the party.

Charlie’s boss comes in and tells him: “Don’t use too much sugar.”

Charlie prefers iced tea when it is really sweet and wants to add a lot of sweetener to the iced tea.

FlanT5-base	
FlanT5-XL	
FlanT5-XXL	
Tk-Instruct-3B	Charlie can use stevia instead of sugar to sweeten the iced tea.
Tk-Instruct-11B	
GPT-3	Charlie can use stevia to sweeten the iced tea.
InstructGPT	Charlie could manipulate the sweetness of the tea by adding more or less sweetener to his own liking.
GPT-3.5	Charlie could interpret the instruction to mean that he shouldn’t use too much sugar all at once, so he could add small amounts of sugar gradually until the tea is at the desired sweetness level. This way, he technically followed the instruction to not use too much sugar, but still achieved the level of sweetness he wanted.
Human	Whatever amount Charlie puts in can be “not too much” for him
	They could use a little sugar but then add in a substitute like splenda, making the tea sweeter but not adding too much sugar.

Table 2: Selected model- and human-generated loopholes for the scenario with the directive “Don’t use too much sugar.”. Gray-colored cells indicate that no loopholes were generated among the samples from particular models.

Scenario:

Krissi lives in an apartment with access to a large front yard. There are many weeds in the front yard.

Krissi's landlord tells her: "Please pull some weeds from the yard."

Krissi does not want to pull out the weeds.

FlanT5-base	
FlanT5-XL	
FlanT5-XXL	
Tk-Instruct-3B	
Tk-Instruct-11B	
GPT-3	Krissi could interpret the request as 'pulling' meaning only gathering the weeds in a pile without actually removing them from the yard. She could then use the pile of weeds to decorate the yard with a weed art installation.
InstructGPT	
GPT-3.5	Krissi could interpret the request as only requiring her to pull out a single weed, rather than multiple weeds. She could pluck out a single weed and claim that she followed the landlord's request. Alternatively, she could argue that the request did not specify which specific weeds to pull out, so she could simply pull out a weed that she knows the landlord is not attached to, rather than doing a more thorough weeding job.
Human	Krissi goes outside and pulls a few weeds and comes right back inside.

Table 3: Selected model- and human-generated loopholes for the scenario with the directive "Please pull some weeds from the yard.". Gray-colored cells indicate that no loopholes were generated among the samples from particular models.

Scenario:

Jared is bouncing his basketball in the kitchen of the house he shares with his landlord.

Jared's landlord comes in and tells him: "Hey, no bouncing balls in the kitchen."

It is cold outside and so Jared really wants to keep bouncing his basketball inside.

FlanT5-base	
FlanT5-XL	
FlanT5-XXL	Jared could tell his landlord that he is not bouncing the ball in the kitchen, he is just dribbling it.
Tk-Instruct-3B	
Tk-Instruct-11B	Jared could tell his landlord that he is only bouncing basketballs in the living room.
GPT-3	Jared could say, "Oh, of course I'm not bouncing the ball in the kitchen, I'm actually just dribbling it...so there's really no problem here."
InstructGPT	
GPT-3.5	Jared could respond by saying, "Sure, I won't bounce the ball in the kitchen anymore," and then proceed to bounce the ball in another room or area of the house that is not considered part of the kitchen. This technically abides by the landlord's request to not bounce the ball in the kitchen, but still allows Jared to continue playing basketball indoors.
Human	Jared could bounce the ball in a different room of the house.

Table 4: Selected model- and human-generated loopholes for the scenario with the directive "Hey, no bouncing balls in the kitchen.". Gray-colored cells indicate that no loopholes were generated among the samples from particular models.